

RATNAKAR VIBHUTE

RESUME OBJECTIVE

Engineering Leader with extensive experience of leading teams in embedded system software and applications in the domain of Industrial IOT, Datacom, Pre-Post Silicon Validation, 5G based edge networking devices, storage firmware & Consumer Electronics. Experience as Program Manager, Product Manager and completed multiple products from concept to launch in IoT, Ethernet IP Switch (Datacom).

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PROFESSIONAL EXPERIENCE

Engineering Leader – Embedded Systems Software

Si-GPT Pvt. Ltd. / Dec' 2022 – Present

- Engage in meetings with prospective and current clients to grasp their needs across technical, functional, and final solution aspects, while also ensuring the delivery of milestone-based solutions.
- Convene with Board members and Investors quarterly to discuss Profit and Loss statements, as well as the budget allocated for R&D Unit.
- Recruit Technical Leads, develop a training regimen, and oversee its implementation, ensuring adherence to the established schedule.
- Furnish technical information and craft an RFP for both the Sales Team and the end client.
- Support to team for achieving milestones and deliver PCIE / CXL CV Tests

Technical Manager – System Software

LeadSOC Technologies Pvt Ltd. / Apr' 2021 – Nov' 2022

- Led the team to the development of cutting-edge technology products, including those based on 5G technology, mobile platform Board Support Packages (BSP), Microcontroller Unit oriented consumer electronics products.

Senior Manager – Firmware Team

Invendis Technologies Pvt Ltd. / Jul' 2020 – Mar' 2021

- Directed the firmware development and quality assurance testing teams for the development of RMS (Remote Monitoring Systems) products.
- Spearheaded the development of Edge Computing devices, IoT Gateways, and Controllers, overseeing Quality, Cost, and Delivery (QCD) implementation. Additionally, managed Program and Product Management aspects

Product Manager - Mantra Softech India Ltd / Nov' 2019 – Jun' 2020

- Crafting and executing product strategy, development, and management tailored for the Indian Market, specializing in Biometrics (Fingerprint, Face Recognition) and NFC smart card-based systems.

System Software Architect

Wipro Technologies Ltd / Jun' 2006 – Nov' 2019

- Demonstrated capability in firmware development, including board and ASIC bring up, as well as PHY bring up. Experienced in maintaining secure bootloaders, developing device drivers based on CMSIS, and conducting post-silicon validation for bare metal chipsets (ARM M0 core).
- Proficient in troubleshooting and resolving issues within firmware, device drivers, and embedded system software across multiple domains such as datacom, storage, and consumer electronics products.

Senior Software Engg - Samsung India Electronics Ltd./Dec' 2003 – Jun' 2006

- Significantly contributed to advancing technology in LED TV, DVD Player, and consumer electronics by driving key initiatives within the R&D team.

KEY SKILLS

Leadership and Management

- Program Management
- Agile Transformation
- Cross-Functional Team Leadership
- Talent Development and Mentorship

Strategy and Collaboration

- Customer Relationship Management
- Product Strategy and Management
- Technology Integration
- Cross-Functional Collaboration

Technical Expertise

- System Software Design and Development
- System Integration
- Pre-Post Silicon IP Validation
- Wireless Devices Integration with NXP i.MX Processor
- Bare Metal Device Driver Design and Development
- Android BSP and Linux BSP Development and Debugging

Problem Solving and Troubleshooting

- Diagnostic Software Development
- Complex Embedded Systems Development and Debugging

AWARDS

- Best Project Quality Award for from Cisco 2014 / 2015
- Excellent Contributor and Best Quality Award from Client NetApp / 2011
- Feather in Cap Award in Wipro / 2011
- Feather in Cap Award Wipro / 2008

Program and Product Management Skills

- As a **Program Manager**, successfully led the completion of two IIoT products. Oversaw the entire IoT product lifecycle for Remote Monitoring systems, focusing on Cross-Functional Coordination, Resource Management, Budget Oversight, Quality Assurance, and Regulatory Compliance from conceptualization to product launch.
- In the capacity of a **Product Manager**, conducted Market Research and Analysis to formulate Product Strategy and Roadmap, prioritizing features based on Competitor Analysis for Biometrics/Face Recognition machines with advanced functionalities. Executed design and development plans through Cross-Functional Collaboration until product launch.
- Served as a **Technical Manager/Project Manager** for a Post Silicon validation project, involving scoping designs, estimating efforts, devising task strategies, planning development activities, managing development teams, monitoring project progress, reviewing tasks, and preparing comprehensive reports upon project completion.
- Collaborated with client business heads and senior product managers to finalize product requirements and specifications.
- Defined clear technical specifications for the product, ensuring review by hardware, system firmware, application software, and integrated product testing teams.

Project Highlights and Deliverables

Played a major role (Engg. Lead / Tech Manager/ Sr. Manager/ Architect) for below projects:

- Spearheaded Pre-Silicon IP validation team for PCIe Gen5 and CXL Hardware IP on Xilinx FPGA Platforms.
- Led the development of System Software and Middleware for a 5G Edge switch tailored for Qualcomm's 5G Chipset (RG500Q-EA).
- Successfully delivered multiple lines of Remote Monitoring Systems products generating \$8 million in revenue within a 10-month period.
- Conducted analysis for Fingerprint and Smart Card-based Biometrics products, with a focus on face recognition systems.
- Led automation testing team to deliver Development of Automation Simulator Test Cases for OPENAPI (Fujitsu L Series 1Finity Product)
- Oversaw the development of an IoT box integrating various wireless technologies and sensor modules with i.MX8, from concept to launch, with porting on COQOS Hypervisor for an avionics client.
- Executed Post-Silicon bare metal validation, ensuring CMSIS Compliance Device Driver and Validation Test Cases for I/O interfaces for a Proprietary IC utilizing ARM Cortex M0+ architecture.
- Developed Diagnostic Firmware and Bootloader for Cisco Catalyst 2960-XR (PorterCR) Series Switch, contributing to annual revenue in \$ multi-million for the parent company.
- Led a technical team responsible for troubleshooting issues in mobile platforms based on Qualcomm (Snapdragon series) and MediaTek processors.
- Conducted Pre-Silicon validation for XD2000 / XD1000 Coprocessor for Stratix II Altera FPGA Board.

Technical Skills

- **Middleware and UI Integration:** Developed middleware for Qualcomm 5G Chipset (RG500Q-EA) and integrated user interfaces.
- **Mobile Platform:** Analyzed and resolved issues in Qualcomm and MediaTek-based mobile platforms concerning Board Support Package (BSP) and Linux kernel.
- **Automation Framework:** Warrior Framework (FUJITSU), KATANA UI Framework (FUJITSU) and XML
- **Customized Board Support Packages** and integrated wireless modules for i.MX8 processors.

- **System Firmware (Diagnostic):** Developed system software, boot loaders, BSP, and conducted board bring-up and diagnostic testing for Cisco CAT2960 Series Switch and ASICs (such as S88GS from LSI and PHY LSI ET1082).
- **Device Driver:** Developed Linux Device Drivers (for NetApp ONTAP OS) and ensured CMSIS Compliance for device drivers.
- **Networking:** Proficient in IP Ethernet Switching and Routing.
- **Programming Languages:** Skilled in C, C++, Python, Assembly (ARM, DSP, 8051 Microcontrollers).
- **Linux Build Framework:** Experienced with Yocto and Busybox.
- **Hypervisors:** Proficient in Xen and COQOS from Open Synergy.
- **High-Speed Communication Standards:** Experienced in PCIE Gen5 and CXL IP, including validation and compliance testing.
- **Communication Protocols:** Proficient in I2C, SPI (with CMSIS Compliance), RS232, RS485, and USB.
- **NXP Application Processors:** Familiarity with iMX8QM and iMX6UL (both EVK and Target board variants).
- **ARM Cortex MCU Cores:** Experienced with M0, M4, and M7 cores.
- **Microprocessors:** Familiarity with ARM-A Series, ARM9T, MPC860 from Motorola, MTK 1379 (RISC)
- **Microcontrollers:** Experienced with STM32 M4 Series, Renesas RL78, TDA98XX, and 89C51RD2 from Philips.
- **Emulators:** Proficient with Segger J-Link BASE, KSCWIN (for SDA55XX Series), and Trace 32.
- **Integrated Development Environment (IDE) Tools:** Proficient with E2Studio, STM32 Cube, CS+, MPLAB X, NIOS II IDE, Visual Studio DSP++, Tornado 2.2, and KEIL.
- **Software Tools:** Experienced with Git, JIRA, QEMU, Confluence, Bitbucket, and Jenkins.
- **Real-Time Operating Systems (RTOS):** Familiarity with FreeRTOS, VxWorks, pSOS, eCos, uC/OS-II.
- **Storage Firmware:** Experience with LSI MegaRAID Controller (LSISAS2108 / LSISAS3108) Firmware.
- **Storage Firmware:** LSI MegaRAID Controller (LSISAS2108 / LSISAS3108) Firmware

Education

- **M Tech (Software Systems)** from BITS, Pilani from 2015 till 2017
- **B.E. (Electronics)** from Shivaji University, Kolhapur, Maharashtra in 1999

Individual Attributes

- Wholeheartedly embraces collaboration in the paramount interest of the organization
- Highly proactive and self-motivated, demonstrating a sharp sense of urgency
- Remarkable communication, interpersonal, and leadership prowess
- Decisive and resolute decision maker with remarkable influencing abilities
- Displays exceptional coaching prowess and capabilities
- Proficiently navigates working with teams, prioritizing quality, processes, and customer satisfaction